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Introduction to Indian Knowledge System

AAS204

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 2 | **Credits:** 2 | **L-T-P:** 2-0-0

Scheme: AKTU B.Tech 1st year, 2026-27 onwards

UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5
Foundations of the Indian Knowledge System	Mathematics and Astronomy	Engineering, Technology and Materials	Health, Environment and Governance	Contemporary Relevance of IKS

Original revision notes written for the AKTU syllabus. This is a first-year course common to all branches. The same course is taught as AAS104 in semester 1 for the other group of students. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the main ideas, people, texts and dates, with comparison tables.
3. **Worked examples**: the small calculations and decodings that can be asked (Sulbasutra √2, Aryabhata's π, katapayadi, Pingala's patterns, Brahmagupta's formula).
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (2 hours):** Unit 1 (meaning and scope, NEP 2020, 14 vidyasthanas, darshanas, oral tradition, pramanas) and Unit 2 (place value, zero, number words, katapayadi, Pingala, Sulbasutras, Aryabhata, Brahmagupta, Bhaskara, Kerala school, Panchanga).
- **Afternoon (2 hours):** Unit 3 (wootz steel, zinc, iron pillar, Harappan planning, Vastu and Shilpa, stepwells, textiles) and Unit 4 (Ayurveda, yoga, environment, Arthashastra, Gita).
- **Evening (1 to 2 hours):** Unit 5 (TKDL, IKS Division, IPR, Panini and NLP, sustainable engineering), then the "Last-hour checklist".
- **In the exam:** write the name of the scholar, the text and the approximate period for every point, and link each idea to a modern engineering or scientific use. That link earns the extra marks.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit. Internal assessment carries another 30 marks.

UNIT 1 Foundations of the Indian Knowledge System

✍️ 2-mark definitions

- **Indian Knowledge System (IKS):** the body of knowledge developed in India over thousands of years in philosophy, science, mathematics, medicine, engineering, arts, language and governance, carried in texts, oral traditions and living practices.
- **Scope of IKS:** knowledge (jnana), science (vijnana) and practice (kala and shilpa); it covers both classical texts and traditional community knowledge (farming, crafts, water management).
- **NEP 2020 and IKS:** the National Education Policy 2020 asks that IKS be part of the curriculum so that students learn India's contributions and connect them with modern knowledge.
- **Vidyasthanas:** the 14 traditional fields of learning: the 4 Vedas, the 6 Vedangas, Purana, Nyaya, Mimamsa and Dharmashastra.
- **Vedangas:** the six "limbs" that support the Vedas: Shiksha (phonetics), Kalpa (ritual procedure, including the Sulbasutras), Vyakarana (grammar), Nirukta (etymology), Chandas (metre) and Jyotisha (astronomy and time-keeping).
- **Darshana:** a school of Indian philosophy (literally "way of seeing").
- **Astika and nastika:** astika schools accept the authority of the Vedas; nastika schools do not.
- **Pramana:** a valid means of knowledge (how we know something is true).
- **Guru-shishya parampara:** the teacher-student lineage through which knowledge was passed on by direct teaching.

💡 Core concepts

The six astika and the main nastika darshanas:

School	Founder or key text	Main idea
Nyaya	Gautama, Nyaya Sutra	Logic and valid reasoning
Vaisheshika	Kanada	Reality made of atoms (paramanu) and categories (padartha)
Samkhya	Kapila	Two principles: purusha (consciousness) and prakriti (matter)
Yoga	Patanjali, Yoga Sutra	Discipline of body and mind for liberation
Mimamsa	Jaimini	Interpretation of Vedic rituals and duty
Vedanta	Badarayana, Brahma Sutra	Nature of Brahman and the self
Charvaka (nastika)	Brihaspati (tradition)	Materialism; only perception is valid
Buddhism (nastika)	Gautama Buddha	Four Noble Truths, Eightfold Path

School	Founder or key text	Main idea
Jainism (nastika)	Mahavira (24th Tirthankara)	Ahimsa, anekantavada (many-sidedness of truth)

The six pramanas:

Pramana	Meaning	Example
Pratyaksha	Perception through the senses	Seeing smoke
Anumana	Inference	Smoke on the hill, so fire on the hill
Upamana	Comparison or analogy	Recognising a wild ox from the description "like a cow"
Shabda	Reliable testimony	A trusted textbook or expert
Arthapatti	Postulation	He is fat but does not eat by day, so he eats at night
Anupalabdhi	Non-perception	No pot is seen on the floor, so there is no pot

- Charvaka accepts only pratyaksha; Nyaya accepts four (the first four); Mimamsa (Bhatta) and Advaita Vedanta accept all six.
- **Link to engineering:** anumana is the basis of scientific reasoning and diagnosis; the Nyaya five-step argument (claim, reason, example, application, conclusion) is like a structured technical proof.

Oral tradition:

- The Vedas were memorised and passed on for centuries before writing, using recitation methods such as **pada, krama, jata and ghana patha**, in which words are chanted in different orders so that any error is caught. This works like an error-checking code.
- UNESCO lists Vedic chanting as an Intangible Cultural Heritage of Humanity.

Key points to remember

- 14 vidyasthanas = 4 Vedas + 6 Vedangas + Purana + Nyaya + Mimamsa + Dharmashastra.
- Six Vedangas: Shiksha, Kalpa, Vyakarana, Nirukta, Chandas, Jyotisha.
- Astika: Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, Vedanta. Nastika: Charvaka, Buddhism, Jainism.
- Six pramanas; Nyaya accepts four, Charvaka one.
- NEP 2020 brings IKS into higher education.

⭐ Important questions

1. Define IKS and explain its scope. Why does NEP 2020 emphasise it?
2. Explain the 14 vidyasthanas and the six Vedangas.
3. Compare the astika and nastika schools of Indian philosophy.
4. Explain the six pramanas with examples and their relevance to scientific reasoning.
5. Explain the oral tradition and the recitation methods used to preserve texts without error.

UNIT 2 Mathematics and Astronomy

✍️ 2-mark definitions

- **Decimal place-value system:** a system in which the value of a digit depends on its place (units, tens, hundreds), using ten symbols including zero. It developed in India and reached the world through Arabic scholars ("Hindu-Arabic numerals").
- **Zero (shunya):** used both as a placeholder and as a number with its own rules. Brahmagupta (628 CE) gave the first rules for arithmetic with zero and negative numbers.
- **Infinity (ananta, kha-hara):** Bhaskara II described a number divided by zero (kha-hara) as unchanged by adding or removing any amount, an early idea of infinity.
- **Bhutasankhya:** a system of writing numbers with words for objects that suggest a number (moon = 1, eyes = 2, Vedas = 4, seasons = 6). Digits are read from right to left. Used in verse to make numbers easy to remember.
- **Katapayadi:** a system that gives each consonant a digit (ka, ta, pa, ya groups), so words and phrases can encode numbers. Digits are read in reverse.
- **Sulbasutras:** texts (Baudhayana, Apastamba, Katyayana; about 800 to 200 BCE) giving geometric rules for building fire altars, including the Pythagorean theorem and an approximation of √2.
- **Panchanga:** the Indian calendar almanac with five elements: tithi (lunar day), vara (weekday), nakshatra (lunar mansion), yoga and karana.

💡 Core concepts

Key mathematicians and astronomers:

Scholar	Period	Text	Main contributions
Pingala	About 3rd to 2nd century BCE	Chandahshastra	Binary-like patterns of short and long syllables; meru prastara (Pascal's triangle)
Baudhayana	About 800 BCE	Baudhayana Sulbasutra	Diagonal rule (Pythagorean theorem), √2 approximation, constructions
Aryabhata	476 to 550 CE	Aryabhatiya (499 CE)	π ≈ 3.1416, sine table, Earth rotates on its axis, eclipses explained by shadows
Brahmagupta	598 to 668 CE	Brahmasphutasiddhanta (628 CE)	Rules for zero and negatives, cyclic quadrilateral area, solutions of quadratic and indeterminate equations
Bhaskara II	1114 to 1185 CE	Lilavati, Bijaganita, Siddhanta Shiromani	Chakravala method, ideas close to differential calculus, arithmetic and algebra textbooks

Scholar	Period	Text	Main contributions
Madhava (Kerala school)	About 1340 to 1425 CE	Works known through later texts	Infinite series for π, sine and cosine (centuries before Europe)
Nilakantha Somayaji, Jyeshthadeva	15th to 16th century	Tantrasangraha, Yuktibhasha	Planetary model; proofs of the infinite series

Pingala's binary idea:

- Each syllable of a metre is laghu (light, L) or guru (heavy, G). A line of n syllables has $\mathbf{2^n}$ possible patterns.
- Listing the patterns in order (prastara) is like counting in binary, with L and G as the two digits.
- The **meru prastara** (Pascal's triangle) gives how many patterns have k heavy syllables.

Madhava series: $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \ldots$ (now called the Madhava-Leibniz series).

Worked examples

Example 1 (Sulbasutra √2). Baudhayana's rule: increase the side by its third, and this third by its own fourth, less the thirty-fourth part of that fourth.

- $\sqrt{2} \approx 1 + 1/3 + 1/(3 \times 4) - 1/(3 \times 4 \times 34) =$ **1.4142157**, against the true value 1.4142136 (correct to 5 decimal places).

Example 2 (Aryabhata's π). "Add 4 to 100, multiply by 8, and add 62,000. This is the approximate circumference of a circle of diameter 20,000."

- $(100 + 4) \times 8 + 62{,}000 = 62{,}832$; $\pi \approx 62{,}832/20{,}000 =$ **3.1416**.

Example 3 (katapayadi). Carnatic music names its 72 melakarta ragas so that the first two syllables give the raga number. For "Mechakalyani", ma = 5 and cha = 6, giving 56; reversed, this is raga number **65**.

Example 4 (Pingala). A line of 4 syllables has $2^4 =$ **16** patterns. Row 4 of the meru prastara is **1, 4, 6, 4, 1**: one pattern with no heavy syllable, four with one, six with two, four with three and one with four (total 16).

Example 5 (Brahmagupta's formula). A cyclic quadrilateral has sides 3, 4, 5 and 6. $s = 9$; area $= \sqrt{(9-3)(9-4)(9-5)(9-6)} = \sqrt{360} =$ **18.97** square units.

Key points to remember

- Place value and zero from India; Brahmagupta's rules for zero (628 CE).
- Bhutasankhya (object words) and katapayadi (consonant digits), both read in reverse.
- Pingala: 2^n patterns, meru prastara.
- Baudhayana $\sqrt{2} = 1.4142157$; Aryabhata $\pi = 3.1416$ (499 CE).
- Bhaskara II (Lilavati, chakravala); Madhava series $\pi/4 = 1 - 1/3 + 1/5 - \ldots$
- Panchanga: tithi, vara, nakshatra, yoga, karana.

⭐ Important questions

1. Explain the Indian contribution of the place-value system and zero, with Brahmagupta's rules.
2. Explain bhutasankhya and katapayadi with examples.
3. Explain Pingala's work on metres and its link to binary numbers and Pascal's triangle.
4. Explain the geometry of the Sulbasutras with the √2 approximation.
5. Write short notes on Aryabhata, Brahmagupta, Bhaskara II and the Kerala school.

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UNIT 3 Engineering, Technology and Materials

✍️ 2-mark definitions

- **Wootz steel:** a high-carbon crucible steel made in South India (and Sri Lanka) from about the 3rd century BCE, exported widely and famous as the raw material of "Damascus" sword blades.
- **Crucible process:** iron and carbon-bearing material (wood, leaves) sealed in clay crucibles and heated for a long time, so that the iron absorbs carbon and melts into uniform steel.
- **Zinc distillation:** at Zawar (Rajasthan), zinc was produced on an industrial scale in the medieval period by downward distillation in clay retorts. Zinc boils at a low temperature (907 °C), so it must be condensed below the furnace instead of tapped as a liquid.
- **Delhi iron pillar:** a wrought-iron pillar about 7.2 m tall and about 6 tonnes, made in the Gupta period (about 4th to 5th century CE), that has hardly rusted in over 1,600 years.
- **Vastu Shastra:** the traditional science of architecture and site planning (orientation, light, ventilation, proportions).
- **Shilpa Shastra:** texts on arts and crafts, including sculpture, iconography and temple building.
- **Stepwell (vav, baori):** a deep well reached by flights of steps, used to store groundwater in dry regions and keep it cool.

💡 Core concepts

Why the iron pillar does not rust: the iron has a high phosphorus content and very little sulphur. Phosphorus helps form a thin, dense, protective passive layer (including iron hydrogen phosphate, also called misawite) on the surface, which stops further corrosion. The dry Delhi climate also helps.

Harappan town planning (about 2600 to 1900 BCE):

- Cities on a **grid** of straight streets crossing at right angles (Mohenjo-daro, Harappa, Kalibangan).
- **Covered drains** along streets with inspection holes; houses connected to them.
- Standardised **burnt bricks** in the ratio 1 : 2 : 4.
- Citadel and lower town; the **Great Bath** at Mohenjo-daro made watertight with bitumen.
- **Lothal** dockyard (Gujarat); **Dholavira** water reservoirs and channels.
- Standard weights and measures.

Water engineering:

- **Stepwells:** Rani ki Vav (Patan, Gujarat, UNESCO World Heritage Site), Chand Baori (Abhaneri, Rajasthan).
- **Tanks and lakes:** temple tanks and village tanks for rainwater harvesting and groundwater recharge.
- **Kallanai (Grand Anicut):** built across the Kaveri by the Chola king Karikala (about 2nd century CE), one of the oldest water-regulating structures still in use.

Temple architecture (Shilpa and Vastu):

- Styles: **Nagara** (north, curved shikhara), **Dravida** (south, pyramidal vimana and gopurams), **Vesara** (a mix, in the Deccan).

- Examples: Brihadeeswara temple, Thanjavur (granite, tall vimana, 11th century); Konark Sun Temple (chariot design with stone wheels used as sundials); Kailasa temple, Ellora (cut from a single rock from the top down).

Textiles:

- Cotton was spun and woven in the Harappan period.
- **Dhaka muslin**: extremely fine cotton cloth.
- **Natural dyeing with mordants** (alum, iron) to fix colours; indigo dyeing.
- **Patola** (Gujarat): double ikat, in which both warp and weft are tie-dyed before weaving.
- **Kalamkari**: hand-painted or block-printed cotton with natural dyes.

Key points to remember

- Wootz: crucible high-carbon steel, Damascus blades.
- Zawar: zinc by downward distillation (zinc boils at 907 °C).
- Iron pillar: Gupta period, 7.2 m, about 6 t, high phosphorus gives a protective layer.
- Harappan: grid, covered drains, bricks 1 : 2 : 4, Great Bath, Lothal dock, Dholavira reservoirs.
- Rani ki Vav, Chand Baori, Kallanai; Nagara, Dravida, Vesara temple styles.

⭐ Important questions

1. Explain the metallurgy of wootz steel and zinc distillation in ancient India.
2. Why has the Delhi iron pillar not rusted? Explain the science behind it.
3. Describe the town planning and drainage of the Harappan civilisation.
4. Explain traditional water management: stepwells, tanks and the Grand Anicut.
5. Explain Vastu and Shilpa Shastra with examples of temple architecture; write a note on Indian textile technology.

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UNIT 4 Health, Environment and Governance

✍️ 2-mark definitions

- **Ayurveda:** the traditional Indian system of medicine ("knowledge of life") focused on balance in the body, prevention, diet and lifestyle.
- **Charaka Samhita:** the classical text on internal medicine (kayachikitsa), covering diagnosis, diet, drugs and the duties of a physician.
- **Sushruta Samhita:** the classical text on surgery (shalya), describing many surgical instruments and procedures such as rhinoplasty (nose reconstruction) and cataract couching. Sushruta is often called the father of surgery.
- **Tridosha:** the three functional principles of the body in Ayurveda: **vata** (movement), **pitta** (metabolism, heat) and **kapha** (structure, stability). Health is their balance.
- **Yoga (Patanjali):** a discipline described in the Yoga Sutras as "yogas chitta vritti nirodhah", the stilling of the fluctuations of the mind.
- **Ashtanga yoga:** the eight limbs: yama, niyama, asana, pranayama, pratyahara, dharana, dhyana and samadhi.
- **Arthashastra:** a treatise on statecraft, economics, law and administration attributed to Kautilya (Chanakya), about the 4th century BCE.
- **Nishkama karma:** the Bhagavad Gita's teaching of doing one's duty without attachment to its results.

💡 Core concepts

Ayurveda in brief:

- Eight branches (ashtanga Ayurveda): internal medicine, surgery, ENT and eye, paediatrics, toxicology, psychiatry, rejuvenation (rasayana) and fertility (vajikarana).
- Prevention through **dinacharya** (daily routine) and **ritucharya** (seasonal routine), diet and good conduct.
- Today, Ayurveda is regulated in India under the Ministry of AYUSH.

Environment in IKS:

- Nature is treated as sacred (rivers, mountains, trees, animals).
- **Sacred groves** (devrai, orans, kavu) protected by communities conserve biodiversity and water sources.
- **Vrikshayurveda:** the traditional science of plant life, covering seed treatment, irrigation and plant health.
- Practices such as the Bishnoi community's protection of trees and wildlife (the Khejarli sacrifice of 1730).

Arthashastra on governance:

- **Saptanga theory:** the seven elements of a state: swami (ruler), amatya (ministers), janapada (territory and people), durga (fort), kosha (treasury), danda (army and justice) and mitra (allies).

- Detailed rules on taxation, trade, weights and measures, labour, forest and mine management, espionage and diplomacy (mandala theory of neighbouring states).
- The welfare of the people is the ruler's duty: "in the happiness of the subjects lies the happiness of the king".

Gita ethics for engineers:

- **Svadharma:** do your own duty well (professional responsibility).
- **Nishkama karma:** focus on doing the work right, not only on rewards.
- **Sthitaprajna:** stay steady in success and failure (stress management).
- **Lokasangraha:** act for the welfare of society.

Key points to remember

- Charaka (medicine), Sushruta (surgery, rhinoplasty, cataract).
- Tridosha: vata, pitta, kapha.
- Ashtanga yoga: yama, niyama, asana, pranayama, pratyahara, dharana, dhyana, samadhi.
- Sacred groves, Vrikshayurveda, Bishnoi (1730).
- Arthashastra by Kautilya; saptanga: swami, amatya, janapada, durga, kosha, danda, mitra.
- Gita: svadharma, nishkama karma, lokasangraha.

⭐ Important questions

1. Explain the principles of Ayurveda (tridosha, prevention) and the contributions of Charaka and Sushruta.
2. Explain Patanjali's ashtanga yoga and its relevance to health and focus for students.
3. Discuss the environmental ethics and practices of traditional India.
4. Explain Kautilya's Arthashastra and the saptanga theory of the state.
5. Explain the ethical teachings of the Bhagavad Gita and their relevance to engineers.

UNIT 5 Contemporary Relevance of IKS

✍️ 2-mark definitions

- **TKDL (Traditional Knowledge Digital Library):** a database set up by CSIR and the Ministry of AYUSH (2001) that records traditional Indian medicinal knowledge in several languages, so patent examiners worldwide can find it as prior art and refuse wrongful patents.
- **Biopiracy:** patenting or commercially using traditional knowledge or biological resources without the consent of, or benefit to, the communities that developed them.
- **IKS Division:** a division set up in 2020 at AICTE under the Ministry of Education to promote research, education and outreach on Indian knowledge systems.
- **Geographical Indication (GI):** an IPR tag for products whose quality comes from their place of origin, such as Darjeeling tea, Banarasi saree and Kanchipuram silk.
- **Panini's Ashtadhyayi:** a Sanskrit grammar (about 5th to 4th century BCE) of about 4,000 short rules (sutras) that generate correct words and sentences from roots and suffixes.
- **Sustainable engineering:** design that meets present needs without harming future generations, using local materials, low energy and recycling.

💡 Core concepts

Patent cases that led to TKDL:

Case	What happened
Turmeric (USA, 1995)	Patent on turmeric for wound healing revoked in 1997 after CSIR showed it was known in old Indian texts
Neem (Europe)	Patent on a neem-based fungicide revoked by the European Patent Office as traditional knowledge
Basmati (USA, 1997)	Most claims of a US patent on basmati rice lines were withdrawn after India's challenge

IPR and traditional knowledge: patents need novelty, so documented traditional knowledge (prior art) blocks false claims. GI tags, the Biological Diversity Act (2002) and benefit-sharing protect communities.

Panini and computing:

- The Ashtadhyayi works like a **formal generative grammar**: a small set of rules, applied in order, with meta-rules for which rule wins in a conflict.
- It uses short technical markers (it-markers) and abbreviations (pratyaharas, such as "ac" for all vowels), similar to variables and character classes in programming.
- Scholars have compared its notation to the **Backus-Naur form** used to define programming languages, and its rule-based analysis is studied in computational linguistics and natural language processing.

Traditional sustainable engineering:

- **Passive cooling:** thick walls, courtyards, jaalis (perforated screens), wind catchers and shaded verandas reduce the need for air-conditioning.
- **Water harvesting:** stepwells, tanks, johads (Rajasthan), kunds and rooftop collection.
- **Local materials:** mud, lime mortar (breathable and self-healing), bamboo and stone with low embodied energy.
- **Circular practices:** repair, reuse and recycling of metals, cloth and organic waste.

IKS today: yoga (International Day of Yoga on 21 June since 2015), Ayurveda and herbal products, millets (International Year of Millets 2023), traditional water systems in modern rainwater-harvesting policies, and IKS-based start-ups.

Key points to remember

- TKDL (2001, CSIR and AYUSH) prevents biopiracy; turmeric (1997), neem, basmati cases.
- IKS Division at AICTE (2020); NEP 2020.
- GI tags: Darjeeling tea, Banarasi saree.
- Panini: about 4,000 sutras, generative rules, compared with Backus-Naur form; used in NLP research.
- Passive cooling, water harvesting and local materials are sustainable lessons for engineers.

⭐ Important questions

1. What is TKDL? Explain how it prevents biopiracy with the turmeric and neem cases.
2. Explain the role of IPR (patents, GI) in protecting traditional knowledge.
3. Explain Panini's grammar and its relevance to computer science and NLP.
4. Discuss traditional sustainable engineering practices and their relevance today.
5. Explain the role of the IKS Division and NEP 2020 in reviving Indian knowledge systems.

Last-hour checklist

☐ IKS meaning and scope; NEP 2020; 14 vidyasthanas; 6 Vedangas; darshana table; 6 pramanas; oral tradition (patha methods).

☐ Place value and zero (Brahmagupta 628 CE); bhutasankhya; katapayadi (raga 65); Pingala 2^n and 1, 4, 6, 4, 1; $\sqrt{2}$ = 1.4142157; π = 3.1416; Brahmagupta area 18.97; Bhaskara II; Madhava; Panchanga.

☐ Wootz; Zawar zinc; iron pillar (phosphorus layer); Harappan planning; stepwells and Kallanai; temple styles; textiles (Patola, Kalamkari, muslin).

☐ Charaka and Sushruta; tridosha; ashtanga yoga; sacred groves and Bishnoi; Arthashastra saptanga; Gita ethics.

☐ TKDL cases; IKS Division (2020); GI and IPR; Panini and Backus-Naur form; passive cooling and water harvesting.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Introduction to Indian Knowledge Systems: Concepts and Applications	B. Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R. N.	PHI Learning	WorldCat
2	Indian Knowledge Systems (Vol. I and II)	Kapil Kapoor, Avadhesh Kumar Singh	D. K. Printworld	WorldCat
3	Pride of India: A Glimpse into India's Scientific Heritage	Samskrita Bharati	Samskrita Bharati	WorldCat
4	Science in India: Past and Present	B. V. Subbarayappa	Nehru Centre, Mumbai	WorldCat

Syllabus source: AKTU B.Tech 1st year CSE stream course structure and syllabus, 2026-27 (official PDF)

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